

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)		The specific heat capacity is the heat required to increase the temperature of 1 kg [or unit mass] of a material by 1 °C (1 degree K – or one degree/ unit temperature rise). [Alternative: equation with all terms fully defined.]	1			1		
	(b)	(i)	Idea: heat lost by boiling water=heat gained by other water (1) $1.6 \times 10^{-3} (100.0 - \theta_f) = 0.6 \times 10^{-3} (\theta_f - 19.5)$ (1) $\theta_f = 78.0 \text{ °C}$ (1) Alternative (for second mark) If $\Delta\theta$ is temperature rise of water in flask $1.6 \times 10^{-3} (80.5 - \Delta\theta) = 0.6 \times 10^{-3} \Delta\theta$ (✓)			3	3	2	
		(ii)	Mass of water = $1.6 \times 10^{-3} \times 1000 = 1.6$ [kg] (1) Application of $m = \rho V$ [even if $0.6 \times 10^{-3} \text{ m}^3$ used $\rightarrow 0.6$ kg] [Can be credited from (b)(i)] Heat lost = $1.6(100.0 - 78.0 \text{ ecf}) \times 4200$ (1) = $1.48 \times 10^5 \text{ J}$ (1) 148 J $\rightarrow$ 1 mark		3		3	3	
		(iii)	[Work done is negligible as] negligible / no change in volume.			1	1		
			Question 3 total	1	3	4	8	5	0